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### Public Resume

# Yang Chong

Computer Vision · Individual Re-Identification · Embedded Sensing | M.S. Candidate

Target roles: Computer Vision Algorithm Engineer / Embedded AI / Edge Deployment Engineer

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## Education

### Hainan University

2024.09 – 2027.06

M.Eng. · New Generation Electronic Information Technology · GPA 3.6/4.0

School of Information & Communication Engineering

Research focus: computer vision, individual re-identification (ReID), vision-language models, embedded intelligent sensing systems

First-Class Scholarship (University Level, Sep. 2026)

12th Datang Cup National Collegiate New Generation ICT Competition — Industry-Education Integration 5G+ Innovation Application Track, Team 3rd Prize (May 2025)

### Sichuan University of Science and Technology

2019.09 – 2023.06

B.Eng. · Electronic Information Engineering · GPA 3.9/4.0

Sichuan Province Outstanding University Graduate, Class of 2023 (Jun. 2023)

National Scholarship (Dec. 2022)

13th LanQiao Cup National Software and IT Professionals Competition — Sichuan Division, MCU Design & Development (University Group), Individual 3rd Prize (May 2022)

National Encouragement Scholarship (Dec. 2021)

# Projects

## Open-World Crimson Snapper Individual Re-Identification via Vision-Language Priors Core R&D

2025.03 – Present

Feature compression and open-set rejection unified on a CLIP-ReID backbone; identity error rate improved by 27%, meeting the single-fish engineering freeze standard

This project delivers a single-fish video runtime system for open-world fish individual re-identification. Centred on track-level identity lifecycle management, five core mechanisms — including Fast Warm-up, post-confirmation identity retention, and dynamic confidence updating — solve stable recognition under track breaks, frame-quality fluctuation, and identity conflicts.

Python · PyTorch · CLIP · ReID · Open-Set Recognition · Feature Compression

## RFID and Multi-View Vision Dual-Modality Crimson Snapper Data Acquisition Device Project Lead

2025.06 – 2026.09

Full-device engineering: from the fish passage structure to multi-threaded synchronized acquisition software

Three 25 fps video streams are captured stably, and acquisition-plus-annotation efficiency improved by 80% over the manual workflow. A reusable engineering build-and-operation standard now supports downstream dataset production.

Embedded Systems · RFID · Global Shutter Cameras · Python Multithreading · USB 3.2 · UPVC Structural Design

## Screening and Standardization of Individual Tagging Methods for Crimson Snapper Experiment Lead

2025.12 – 2026.08

A controlled multi-method experiment driving tagging standardization and selection guidance

Dorsal injection marking survival for adult fish improved to 80%; a tagging-method selection guide by fish size class was produced, providing a stable marking basis for subsequent individual-recognition dataset construction.

Experimental Design · Control Experiment · Standardized Workflow · Animal Tagging Techniques · Data Consolidation

Core Skills

Programming Languages

Python · C/C++ · VBA batch data processing · 8051 / STM32 microcontrollers · Multi-threaded programming

Frameworks & Tools

PyTorch · OpenCV · ONNX model conversion · FFmpeg · Linux systems · Edge deployment · Serial communication · USB bus protocols (USB 3.2) · RFID systems (134.2 kHz) · Analog and digital circuit debugging · Full-device assembly and commissioning · Mechanical structure design · Industrial-grade waterproof enclosure · AutoCAD structure design · Microsoft Office suite

Research Directions

CNN · Transformer · CLIP vision-language models · Individual re-identification (ReID) · Open-set recognition · Feature compression · Image classification

Honors & Awards

|                                                                         |         |
|-------------------------------------------------------------------------|---------|
| <b>First-Class Scholarship (University Level)</b>                       | 2026.09 |
| Hainan University                                                       |         |
| <b>Sichuan Province Outstanding University Graduate (Class of 2023)</b> | 2023.06 |
| Sichuan Provincial Education Department                                 |         |
| <b>National Scholarship</b>                                             | 2022.12 |
| Ministry of Education, PRC                                              |         |
| <b>Outstanding Student Cadre</b>                                        | 2022.11 |
| Sichuan University of Science and Technology                            |         |
| <b>Outstanding Innovation Trainee</b>                                   | 2022.07 |
| Sichuan University of Science and Technology                            |         |
| <b>National Encouragement Scholarship</b>                               | 2021.12 |
| Ministry of Education, PRC                                              |         |
| <b>Three Good Student</b>                                               | 2021.11 |
| Sichuan University of Science and Technology                            |         |
| <b>First-Class Scholarship (University-level)</b>                       | 2021.11 |
| Sichuan University of Science and Technology                            |         |
| <b>May Fourth Red Banner Pacesetter</b>                                 | 2021.05 |
| Sichuan University of Science and Technology                            |         |
| <b>Advanced Individual in Military Training</b>                         | 2021.03 |
| Sichuan University of Science and Technology                            |         |
| <b>Certificate of Completion, "Young Marxist Training Programme"</b>    | 2020.12 |
| Sichuan University of Science and Technology                            |         |

An Intelligent Classroom Attendance System

Utility Model Patent

Second Inventor

ZL 2022 2 0475134.9

Granted in July 2022

Undergraduate Engineering Innovation